OMAR CUSMA FAIT



Data Science and Artificial Intelligence Student at the University of Trieste

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ABOUT ME

I possess a strong background in **mathematics** and **physics**, which has refined my **analytical thinking** and **problem-solving** skills. These abilities have proven invaluable in approaching complex challenges from a logical perspective.

WORK EXPERIENCE

Math Teacher

ISIS Gregorčič

March 2025 – June 2025 | Gorizia

- Taught mathematics at the high school level, preparing and delivering lessons, supporting students in building solid problem-solving foundations, and assessing progress across the school term.

INTERESTS

Python | Mathematics | Physics | Science | Simulations | Reinforcement Learning | AI | Machine Learning | Big Data | Deep Learning

REFEREES

Dr. Elena Orlando
INAF
orland.ele@gmail.com

Peter Černic
Principal, ISIS Gregorčič – Gorizia
Available upon request

Marco Zennaro
Internship supervisor
Available upon request

Alessio Ansuini
Thesis supervisor
Available upon request

DIGITAL SKILLS

Python ●●●●◐

Pandas | NumPy | MatPlotLib | GitHub | PyCharm | Latex | Krita | Windows | Excel | Word | Pascal | PowerPoint

LANGUAGES

Italian (native) ●●●●●

Slovenian (native) ●●●●◐

English (B2) ●●●●◐

Learning: Japanese

EDUCATION

Master's Degree in Data Science and Artificial Intelligence

Università degli Studi di Trieste – DSAI

Thesis title: Unsupervised State Bucketing for Mixture of Experts in Resource-Constrained Chess Engines

Bachelor's Degree in Physics

Università degli Studi di Trieste

Thesis title: A Comparative Analysis of Neural Network Methods for Source Detection

High School Diploma Applied Sciences

Istituto di Istruzione Superiore Gabriele D'Annunzio

PROJECTS

Astrophysics Internship

2023 INAF

- In a Study of the local correlation between two HEALPix images, I faced an open-ended problem for which I had to come up with my own solution to the problem.

AI internship

2026 ICTP

- I developed a data-driven Mixture-of-Experts architecture to optimize resource-constrained NNUE chess engines through gradient-informed sample clustering and a lightweight inference dispatcher.

Semantic Retrieval System with LLMs

- A system that can search through a folder of documents to find the most relevant one, highlight the most relevant text in the document, or group the files in folders based on meaning.

Royal Game of Ur Self-Reinforcement Learning AI

- I applied my knowledge in Reinforcement Learning by creating a bot that learns to play well at the game through self-play only.

Wikipedia Recommendation System

- With my team, we created an online platform that recommends Wikipedia pages based on your preferences. I later improved the system by leveraging a lightweight sentence embedding model.

Quantum Ground State Simulation

- I applied my knowledge in physics to write a program to find the ground state of a quantum system of interacting particles.

Peak Detection in Noisy Signal

- I helped a friend in automating the detection and localization of peaks in noisy spectroscopic data on a user-friendly interface.

tisext – Intelligent Time Series Extension

- A Python library that **extends periodic time series** by learning underlying patterns, even in noisy or interrupted signals. I tested it on a real-world dataset to **forecast product demand**.

Telegram Bot

- I created a general-purpose bot that manages and interacts with Telegram users in real time.

Mini-Hold'Em

- I understood and implemented **Counterfactual Regret Minimization (CFR)** to create a strong MiniPoker bot, despite hidden information playing a significant factor.

Mancala Self-Reinforcement Learning AI

- A self-reinforcement learning AI for the game of Mancala. The agent uses 2-ply exhaustive search and a **neural network-based value function** to choose the best move. The value function leverages self-play, and is trained from scratch.